

**How many hours a year does a researcher
spend on each research task?**

Two types of tasks

- **Paper tasks** are done to produce particular papers.
 - collecting a dataset, running the estimation, drafting the manuscript
- **General tasks** are ongoing and not tied to particular papers.
 - teaching a course, supervising students, reviewing for journals

Paper tasks

hours per year = hours per instance $\times$ instances per paper $\times$ papers per year

Paper tasks: the components

$$\text{hours per year} = \text{hours per instance} \times \text{instances per paper} \times \text{papers per year}$$

$$\text{hours per instance} = \text{team hours one instance takes} \times \text{share not counted under another task}$$

$$\text{instances per paper} = \text{share of papers that involve the task} \times \frac{\text{instances per paper that involves it}}{\text{papers per instance}}$$

$$\text{papers per year} = \frac{\text{papers a researcher publishes per year}}{\text{authors per paper}}$$

Hours per instance

hours per instance = team hours one instance takes $\times$ share not counted under another task

Team hours one instance takes

- Estimated by Claude Opus 4.8 from each task's substeps.
- Validated against lab protocols from protocols.io.

Share not counted under another task

- Determined by Claude Opus 5.5: it flags pairs of tasks whose steps contain the same work and counts that work under one of them, the keeper. The other task loses the share of its hours that is the shared work.
- 41% of paper tasks lose some work this way; the median one keeps 60% of its hours.
- Example: *Decompose wage gaps between groups* and *Decompose aggregate outcomes into components* both run Oaxaca–Blinder decompositions; the second keeps 30%.

Instances per paper

$$\text{instances per paper} = \text{share of papers that involve the task} \times \frac{\text{instances per paper that involves it}}{\text{papers per instance}}$$

Share of papers that involve the task

- About 100 papers sampled per subfield; Claude Sonnet 5 reads each and judges whether it involves the task.
- Median 18% of papers (IQR 7–41%).

Instances per paper that involves it and papers per instance

- Determined by Claude Opus 5.5 from each task's description and substeps.
- Instances per paper: usually 1 (67% of tasks). *Respond to peer reviewers*: 2, two rounds per paper.
- Papers per instance: above 1 when one instance feeds several papers. *Harmonize occupation and industry codes across data sources*: 1.5, since a harmonized crosswalk is reused in later papers. Below 1 when some instances never become a paper.

Papers per year

$$\text{papers per year} = \frac{\text{papers a researcher publishes per year}}{\text{authors per paper}}$$

- From OpenAlex, journal articles published 2015–2019 in each subfield.
- Medians across subfields: 2.0 papers per year and 3.1 authors per paper. Authors per paper range from 1.1 (Literature) to 5.9 (Biomedical Sciences).

General tasks

$$\text{hours per year} = \text{hours per instance} \times \text{instances per year}$$

$$\begin{aligned} \text{hours per instance} &= \text{hours the researcher spends per instance} \\ &\quad \times \text{share not counted under another task} \end{aligned}$$

$$\text{instances per year} = \text{share of researchers who do it} \times \text{instances per year for each of them}$$

Instances per year

instances per year = share of researchers who do it $\times$ instances per year for each of them

Share of researchers who do it

- Estimated by Claude Opus 5.
- Median 92% for general tasks, 90% for subfield-level tasks.

Instances per year for each of them

- Determined by Claude Sonnet 5.5, with the task's substeps shown.
- Median 2 a year. *Evaluate manuscripts for journals* in Labor Economics: 95% review, 6 manuscripts each, so $95\% \times 6 = 5.7$ reviews per researcher per year.

Worked example: a paper task in Labor Economics

Respond to peer reviewer comments and revise manuscripts for publication

One round of responses and revisions takes the team 121 hours, none of it counted under another task. 72% of labor economics papers go through revision, with two rounds each. A labor economist publishes 2.0 papers a year with 2.04 authors per paper.

$$\text{hours per instance} = 121 \times 100\% = 121$$

$$\text{instances per paper} = 72\% \times 2 \div 1 = 1.44$$

$$\text{papers per year} = 2.0 \div 2.04 = 0.98$$

$$\text{hours per year} = 121 \times 1.44 \times 0.98 \approx 171$$

Worked example: a general task in Labor Economics

Supervise graduate students and postdoctoral researchers on research projects

Supervising one student for a year takes 60 hours, none of it counted under another task. 89% of labor economists supervise, three students each.

$$\text{hours per instance} = 60 \times 100\% = 60$$

$$\text{instances per year} = 89\% \times 3 = 2.7$$

$$\text{hours per year} = 60 \times 2.7 \approx 160$$